

EDUCATION

McGill University

Sept. 2017 to May. 2022

B.Sc. Computer Science

Minor in French

SKILLS

PROGRAMMING LANGUAGES: Python, C/C++, Ruby, JavaScript

TECHNOLOGIES: Claude, MCP, Linux, Bash, Git, SQL, NoSQL, REST, GraphQL, HTML/CSS, React, Rails, Django, Pandas, PyTorch, Google Cloud, AWS, Docker

LANGUAGES: English, French, Turkish

EMPLOYMENT

ARC GROUP BENEFITS INC. · Software Developer · Montréal, QC 2023 Sept. to Present

- Led the migration of operational teams from waterfall-style project tracking to Asana by building a bidirectional Rails–Asana integration using the Asana API and webhooks, enabling agile workflows and automating downstream actions such as policy renewals and report generation
- Designed a concurrency-control mechanism to serialize webhook processing around Rails transactions, preventing race conditions and ensuring webhook-driven updates were queued and applied after record locks were released
- Shipped a Rails commission platform handling configurable, multi-party payout splits with per-recipient, per-category rate rules, including reconciliation logic when a third party withholds the full payment upfront and remits the remaining share separately, and cron-scheduled statement generation, replacing a manual process
- Served as engineering escalation point for 80+ production incidents and bugs, including recovery from a migration failure that wiped customer data, restoring from a prior state and verifying integrity via system logs
- Strengthened SOC 2 and security compliance by closing admin access-control gaps, restricting sensitive reports by role, automating contract-end account revocation, and eliminating user-enumeration and information-leakage vectors in UI
- Reviewed 300+ pull requests across nine engineers, catching defects before release and keeping conventions consistent

AMAZON MUSIC · Software Development Engineer · Montréal, QC June 2023 to Aug. 2023

- Ported Swift/Kotlin screens to React Native, consolidating iOS/Android into one codebase and reducing maintenance
- Built the customer onboarding interface with scrollable lists, carousels, and interactive navigation controls
- Connected components to GraphQL, rendering live account and catalogue data with loading and error states

SHOPIFY · Backend Developer · Montréal, QC June 2022 to May 2023

- Built and scaled custom data (metafields and metaobjects) in Rails for Shopify's 1st- and 3rd-party apps, migrating endpoints from REST to GraphQL, replacing deprecated content permissions with metaobjects
- Reduced metaobject create/update latency by 10% through bulk operations; hardened the Metafield loader with fallbacks and resolved a Rails/MySQL string-encoding mismatch
- Implemented country-specific currency localization for money metafields in the Shopify's admin API and extended GraphQL schemas with new fields to eliminate redundant client-side queries
- Diagnosed cross-API inconsistencies through targeted debugging and contributed fixes across services; maintained Shopify's public API documentation with validated GraphQL query examples

MATROX · Software Designer · Dorval, QC Jan. 2022 to April 2022

- Worked on the development of the Mura IPX video capture cards on Windows and Linux platforms using C/C++
- Integrated the SRT protocol into Matrox's Vwlib library, cutting down the packet loss of streams to less than 1%
- Refactored the codebase of casting protocols in C++, delivering a clean interface for a complex subsystem
- Added timestamps on system outputs of the encoder/decoder to pinpoint behaviour changes during debugging

PROJECTS

COMPUTER VISION PROJECT Aug. 2026 to Present

- Built a machine learning model that detects plastic residue on water bodies using instance segmentation
- Generated synthetic training images with diffusion inpainting, replacing cut-and-paste composites
- Refined masks with SAM 2 and trained YOLO11 segmentation on native polygon labels in PyTorch
- Repackaged the workflow as a pip-installable CLI (synthesize, train, infer) with modern Python tooling

SLEEP LOG ANALYSIS TOOL May 2022 to Sept. 2026

- Converted McGill's Attention, Behaviour and Sleep Laboratory's survey data into SPSS variables in Python
- Rebuilt the monolithic notebook as an installable package with CLI commands and typed modules

- Added a seeded synthetic data generator, so the pipeline runs end-to-end with example results
- Transposed horizontally-integrated data for faster processing using NumPy and pandas